MATH 2306 - Differential Equations Course Project

Undergraduate Research Experience Through Mathematical Modeling

Project Weight: 20% of the Course Grade

1 Purpose of the Project

The purpose of this project is to give you an introductory experience with **undergraduate research using differential equations**.

In regular homework problems, the differential equation is usually given to you. In research and applied mathematics, the process often begins much earlier. You must ask a question, decide what quantities should be modeled, make reasonable assumptions, formulate a differential equation or a system of differential equations, analyze the model, perform simulations, and interpret what the mathematics says about the original problem.

For this project, you will independently investigate an application of differential equations. The project is designed to be challenging but accessible to students who are still early in their undergraduate studies. You are **not** expected to produce a publishable research paper or prove advanced theorems. Instead, the goal is to experience the main stages of the research process in a mathematically meaningful and organized way.

By completing the project, you should gain experience with the following research skills:

- identifying a focused research question;
- reading and summarizing mathematical or scientific background sources;
- translating assumptions into differential equations;
- defining variables, parameters, units, and initial conditions;
- performing analytical or qualitative mathematical analysis;
- using numerical simulation to investigate model behavior;
- comparing different parameter values or scenarios;
- interpreting mathematical results in the context of the application;
- discussing limitations of a mathematical model;
- communicating mathematical work in written and oral forms.

2 Project Components

The project has two required components:

1. **Written Research Report**
2. **Recorded Video Presentation**

The base project score is 100 points:

Written Report: 70 points   Video Presentation: 30 points.

An optional research poster can earn **10 bonus points** on the project score. For example, a base project score of 92/100 becomes 102/100 after receiving the poster bonus.

3 Choosing a Research Topic

You may select one of the instructor-supported research directions listed below, or you may propose your own topic.

If you choose your own topic, you **must consult with the instructor and receive approval** before beginning the full project. This consultation is intended to make sure that the topic

- has differential equations as a central mathematical component;
- is appropriate for the level of the course;
- has a focused question that can reasonably be studied during the semester;
- has sufficient background information or data available;
- is not so broad that meaningful mathematical analysis becomes impossible.

A topic does not have to be completely new to mathematics. Undergraduate research often begins by studying an existing model, reproducing part of its behavior, changing an assumption, comparing parameter values, or applying the model to a new situation.

4 Instructor-Supported Research Directions

The following topics are connected to mathematical-modeling research conducted by the instructor and collaborators. The published studies are substantially more advanced than what is required for this course. Your project should use a **simplified version** appropriate for an introductory differential equations course.

4.1 Topic A: Smoking Dynamics and Health Education

Smoking behavior can be studied with compartmental differential-equation models in which individuals move among behavioral states such as nonsmokers, smokers, former smokers, or health-educated individuals. A course-level project might investigate a simplified model with an education or prevention mechanism.

Possible research questions include:

- How does increasing the health-education rate affect the long-term smoking population?
- How does the smoking-initiation rate change the equilibrium or peak number of smokers?
- What happens when relapse is included in the model?
- How do model predictions differ with and without an education intervention?
- Which parameters appear to have the strongest effect on smoking prevalence?

Students are not expected to reproduce advanced threshold, stability, or bifurcation proofs from the published research. Equilibria, numerical solutions, parameter comparisons, and careful interpretation are appropriate goals.

4.2 Topic B: Meme and Information Spread Dynamics

Internet memes and other forms of online information can spread through a population in ways that resemble epidemic transmission. A simplified model may classify users as

$$S(t) = \text{users who have not yet encountered the meme,}$$

$$I(t) = \text{users currently sharing or spreading the meme,}$$

and

$$R(t) = \text{users who are no longer actively sharing it.}$$

A basic project may begin with an SIR-type system. A more ambitious project could introduce a return or "reactivation" mechanism to represent renewed interest.

Possible research questions include:

- How does the sharing rate affect the peak number of active spreaders?
- How does the loss-of-interest rate affect the lifetime of a meme?
- Can a simple reactivation mechanism create a second popularity peak?
- How do different parameter choices produce rapid decay, slower decay, or long-term persistence?
- Can a model trajectory be compared with a Google Trends popularity curve?

The project should focus on differential-equation modeling; advanced stochastic differential equations or machine-learning methods are optional and are **not required**.

4.3 Topic C: Language Competition and Language Shift

Differential equations can be used to model competition between languages spoken in the same population. A simple model may use

$$x(t) = \text{fraction of the population speaking Language 1,}$$

$$y(t) = \text{fraction of the population speaking Language 2,}$$

with

$$x(t) + y(t) = 1.$$

More advanced versions may include a third group of bilingual speakers.

Possible research questions include:

- How does the prestige or attractiveness of a language affect long-term language use?
- Under what parameter choices does one language dominate?
- Can both languages coexist in the model?
- How does allowing bilingual speakers change the predicted outcome?
- How sensitive are long-term predictions to the initial fractions of speakers?

This topic provides a good opportunity to study nonlinear differential equations, equilibria, parameter sensitivity, and model assumptions.

4.4 Topic D: COVID-19 or Other Infectious-Disease Modeling

Epidemic models use systems of differential equations to represent movement among disease states. A basic starting point is an SIR model:

$$\boxed{\begin{aligned} S' &= -\beta\frac{SI}{N}, \\ I' &= \beta\frac{SI}{N} - \gamma I, \\ R' &= \gamma I. \end{aligned}}$$

A project can then add one carefully chosen feature such as vaccination, reduced transmission due to social measures, or an exposed class.

Possible research questions include:

- How does the transmission rate affect the size and timing of the infection peak?
- How does vaccination change the peak number of infected individuals?
- How does reducing contact or transmission change the epidemic curve?
- What is the effect of changing the recovery rate?
- How do two intervention scenarios compare?

The purpose is mathematical modeling, not medical advice. Any interpretation should remain within the assumptions and limitations of the model.

4.5 Topic E: Student-Proposed Application

You may choose another application that interests you. Possible examples include predator–prey or ecological competition, invasive species, rumor spread, chemical reactions, pharmacokinetics, population harvesting, heat transfer, electrical circuits, traffic or transportation, environmental models, economic or social dynamics, or another approved application involving differential equations.

A student-proposed topic requires consultation and approval from the instructor.

5 The Research Process

A strong project should follow a process similar to the following.

5.1 Step 1: Formulate a Focused Question

Avoid topics that are only broad subjects such as "COVID-19" or "population growth." Instead, ask a question that the model can investigate. For example:

> How does increasing the vaccination rate change the peak number of infected individuals in a simplified epidemic model?

or

> How does the health-education rate affect the long-term proportion of smokers in a simplified smoking model?

5.2 Step 2: Study the Background

Find credible sources that explain the application and, when possible, existing mathematical models. Your final report should normally contain at least **five credible references**, including at least **three scholarly source** such as a peer-reviewed paper, scholarly book, or instructor-provided research article.

5.3 Step 3: Define the Model

Clearly define all state variables, parameters, units when appropriate, initial conditions, and assumptions. Then derive or justify each differential equation. Do not simply display a system of equations without explaining where its terms come from.

5.4 Step 4: Analyze the Mathematics

The appropriate analysis depends on the model. Possible approaches include solving the differential equation analytically, finding equilibria, determining where solutions increase or decrease, discussing stability, constructing phase lines or phase-plane plots, investigating conservation laws, comparing long-term behavior, or determining thresholds when appropriate.

You are not required to use every technique above. Use the mathematics that is meaningful for your particular model.

5.5 Step 5: Perform Numerical Simulations

Use computational tools to investigate the model. Python/Jupyter notebooks are recommended, although another instructor-approved computational tool may be used.

Your report should include at least **two meaningful simulation figures or computational comparisons**. Useful investigations include comparing parameter values, comparing intervention scenarios, studying initial conditions, plotting multiple state variables, constructing a phase-plane trajectory, or comparing model predictions with data.

5.6 Step 6: Interpret and Critique the Model

Discuss what the results mean in the application, whether they answer the research question, which parameters matter, which assumptions may be unrealistic, what the model omits, and how the model could be improved.

6 Written Report Requirements

The written report should be approximately **5–8 pages**, excluding references and optional appendices. Quality is more important than reaching a particular page count.

The report should use a short research-paper style and contain:

1. Title
2. Abstract (approximately 100–150 words)
3. Introduction and Research Question
4. Background
5. Model Formulation
6. Mathematical Analysis
7. Numerical Simulations and Results
8. Discussion and Model Limitations
9. Conclusion
10. References

An appendix containing computational code may be included if useful. All borrowed ideas, models, equations, data, figures, and factual claims must be cited appropriately.

7 Recorded Video Presentation

Submit a **10–15 minute recorded presentation** summarizing your project. The presentation should include the application and motivation, research question, variables and parameters, differential equation or system, model construction, main mathematical analysis, important simulation figures, findings, limitations, and possible future work.

The presentation should not consist of simply reading the written report. Explain the mathematics in your own words and guide the audience through the main ideas.

8 Optional Research Poster Bonus

Students may prepare an additional **research poster** summarizing the project. A complete and professional poster that accurately presents the research question, model, mathematical analysis, simulations, conclusions, and references will earn

$$\boxed{+10 \text{ bonus points on the project score.}}$$

For example,

$$92/100 \longrightarrow 102/100.$$

The poster should be designed for a mathematical or undergraduate research poster session, not as a copy of the written report.

9 Project Milestones

Milestone	**Due Date**
Topic selection / instructor consultation	09/17
Short topic proposal and research question	09/24
Model formulation / progress check	10/08
Simulation / analysis progress check	10/22
Final written report	11/19
Recorded presentation	11/19
Optional research poster	11/19

10 Starting References for Instructor-Supported Topics

The following papers are possible starting points. Students are not expected to understand every mathematical detail.

1. P. Xiao, Z. Zhang, and X. Sun, "Smoking dynamics with health education effect," *AIMS Mathematics*, 3(4), 584–599, 2018. DOI: 10.3934/Math.2018.4.584.

2. P. Xiao and B. Wood, "Social Reinforcement in Age-Structured Smoking Dynamics: The Role of Education and the Allee Effect," *Mathematics*, 14(13), 2271, 2026. DOI: 10.3390/math14132271.

3. A. Lonnberg, P. Xiao, and K. Wolfinger, "The growth, spread, and mutation of internet phenomena: A study of memes," *Results in Applied Mathematics*, 6, 100092, 2020. DOI: 10.1016/j.rinam.2020.100092.

4. W. Little and P. Xiao, “Temporal pattern classification of internet meme propagation: A hybrid machine learning approach,” *Electronic Research Archive*, 33(9), 5323–5346, 2025. DOI: 10.3934/era.2025238.

5. C. Sutantawibul, P. Xiao, S. Richie, and D. Fuentes-Rivero, “Revisit Language Modeling Competition and Extinction: A Data-Driven Validation,” *Journal of Applied Mathematics and Physics*, 6, 1558–1570, 2018. DOI: 10.4236/jamp.2018.67132.

6. G. Young, P. Xiao, K. Newcomb, and E. Michael, “Interplay between COVID-19 vaccines and social measures for ending the SARS-CoV-2 pandemic,” *F1000Research*, 10:803, version 2, 2022. DOI: 10.12688/f1000research.54729.2.

Additional publications can be located through the instructor’s Google Scholar profile:

`https://scholar.google.com/citations?user=OHe_hWgAAAAJ&hl=en`